



First Semester Examination 2024/2025 Academic Session

Instruction: Answer question one (1) and any other two (2) questions.

Course Code: ORC 301/309

Code Title: Operations Research

Unit: 2

Time allowed: 2 hours

1a. Briefly discuss the history of Operations Research (6 Marks)

1b. In a tabular form compare PERT and CPM (8 Marks)

1c. Solve the transportation problem given in the table below using the Northwest Corner Method (10 Marks)

(Note the cost of transportation is in Naira)

	1	2	3	4	Supply
1	10	2	20	11	15
2	12	7	9	20	25
3	4	14	16	18	10
Demand	5	15	15	15	

2a. Define the following terms (2 Marks Each)

- Pivot Element
- Feasible Solution
- Optimal Solution
- Objective Function
- Degenerative Solution

2b. Every shift Resident Doctor in Federal Medical Centre Mubi works for five (5) consecutive day and two (2) consecutive off. The Five (5) days of work can start on any day of the week and the scheduled of off relatives

indefinitely. The Medical Centre needs the following minimum number of Doctors to work on a given day as shown in the table below

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
35	55	60	50	60	50	45

No more than forty (40) Doctors can start their five (5) consecutive working days on the same day. Formulate this problem as a Linear Programming Model to Minimize the number of Doctors to be recruited in Federal Medical Centre Mubi (5 Marks)

2c. Solve this Linear Programming problem using Graphical Method (8 Marks)

Max (Z) = $2X_1 + X_2$ such that

$$X_1 + 2X_2 \leq 10$$

$$X_1 + X_2 \leq 6$$

$$X_1 - X_2 \leq 2$$

$$X_1 - 2X_2 \leq 1$$

$$X_1, X_2 \geq 0$$

3a. The physical and planning development unit of Adamawa State University Mubi is planning to embark on a project for the construction of New Faculty building for the Faculty of Computing. The activity time and sequence required to undertake the project is given below (10 Marks)

Activity	Immediate Predecessor	Time (Weeks)
A	-	3
B	A	2
C	A	4
D	A	4
E	B	6
F	C, D	6
G	D, F	2
H	D	3
I	E, G, H	3

Draw the Network diagram of the project showing the weights and directions

3b. Using Simplex Method solve this Linear Programming Model (Just the Initial Tableau and First Iteration) (10 Marks)

Max (Z) = $23X_1 + 32X_2$ such that

$$10X_1 + 6X_2 \leq 2500$$

$$5X_1 + 10X_2 \leq 2000$$

$$X_1 + X_2 \leq 500$$

$$X_1, X_2 \geq 0$$

3c. List the three (3) phases of Project Management (3 Marks)

4a. What is Decision? (5 Marks)

4b. Discuss the following terms (3 Marks Each)

- i. Decision Making Under Certainty (DMUC)
- ii. Decision Making Under Uncertainty (DMUU)
- iii. Decision Making Under Risk (DMUR)

4c. List any Four (4) Properties of Linear Programming Model (8 Marks)

5a. Explain the following terms (2 Marks Each)

- i. Surplus Variable
- ii. Slack Variable
- iii. Simplex Method
- iv. Competitive Situation
- v. Competitive Strategy
- vi. Saddle Point

5b. Formulate the initial Tableau for the following Linear Programming Model (11 Marks)

Min (Z) = $3X_1 + 2.5X_2$ such that

$$2X_1 + 4X_2 \geq 40$$

$$3X_1 + 2X_2 \geq 50$$

$$X_1, X_2 \geq 0$$



ADAMAWA STATE UNIVERSITY, MUBI

FACULTY OF SCIENCE

DEPARTMENT OF COMPUTER SCIENCE

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Course Code: ORC 301

Code Title: Operations Research

Unit: 2

Time allowed: 2 1/2 hours

Instruction: Answer question one (1) and any other two (2) questions.

- (a) Explain any four features of Operations Research (4mks)
(b) Discuss the scope of Operations Research applications (8mks)
(c) Discuss any four assumptions of linear programming models (4mks).
(d) Determine the initial basic feasible solution of the LP model below using Simplex method.

$$\begin{aligned} \text{Max } Z &= 5x_1 + 4x_2 \\ \text{Subject to: } &6x_1 + 4x_2 \leq 10 \\ &x_1 + 2x_2 \leq 6 \\ &-x_1 + x_2 \leq 1 \\ &x_2 \leq 2 \\ \text{Where } &x_1, x_2 \geq 0 \end{aligned}$$

(8mks)

- (a) What is the limitation of graphical method over simplex method? (4mks)
(b) Discuss any three (3) components of a Linear Programming model. (9mks)
(c) Use the graphical method to solve the LP model below:

$$\begin{aligned} \text{Max } Z &= 2x_1 + x_2 \\ \text{Subject to: } &x_1 + 2x_2 \leq 10 \\ &x_1 + x_2 \leq 6 \\ &x_1 - x_2 \leq 2 \end{aligned}$$

Where $x_1, x_2 \geq 0$

(10mks)

- (a) Show the generalized structure of a Linear Programming (LP) model. (7mks)
(b) ADSU farm is engaged in breeding cows. The cows are fed on various products grown on the farm. Because of the need to ensure certain nutrients, it is necessary to buy additional one or two products, which we shall call A and B. the nutrient constituents (vitamin and minerals) in each unit of the products are given below:

Products	Nutrients		
	1	2	3
A	40	6	15
B	8	10	12
Amount of minimum nutrients	108	40	110

If product A costs N20/unit and product B costs N40/unit. Formulate the LP model for products A and B to be purchased at lowest possible cost so as to provide the cows nutrients not less than that given on the table. (7mks)

© Discuss the following:

- i. Decision under risk (3mks)
 - ii. Decision under certainty (3mks)
- Decision under uncertainty. (3mks)

4. (a) Differentiate between CPM and PERT, the use of table will attract more marks. (12mks)

(b) Draw the network diagram of a marriage project below and identify the critical path/activity. (11mks)

Activity	Immediate Predecessor	Time (weeks)	Activity Description
A	-	3	Getting someone you love
B	A	2	Prepare for dinner party
C	A	4	Prepare for Engagement
D	B	4	Announce to public
E	C,B	6	Involve parents for background checks
F	E	6	Mobilize resources to pay bride price
G	D,F	2	Prepare for wedding
H	G	3	Post wedding visitations and honey moon

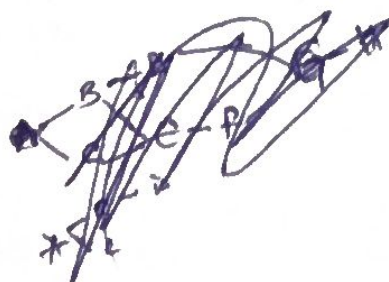
5. The table below summarizes a transportation parameters of a Transportation Problem (TP).

	A	B	C	D	E	Supply
1	13	33	31	8	20	15
2	14	9	17	6	10	10
3	25	11	12	17	15	12
4	10	21	13	36	17	20
Demand	10	20	10	10	7	

Determine the initial basic feasible solution using:

- i. The least cost method (9mks)
- ii. The North-west corner method (9mks)

(b) Which method do you think is appropriate to use and why? (5mks)



ORC 301 Test. Instructions: Answer all questions

1. (a) What is the limitation of graphical method over simplex method?

(2 marks)

(b) Use the graphical method to solve the LP model below

$$\text{Max: } Z = 2x_1 + x_2$$

$$\text{subject to: } x_1 + 2x_2 \leq 10$$

$$x_1 + x_2 \leq 6$$

$$x_1 - x_2 \leq 2$$

$$\text{where } x_1, x_2 \geq 0$$

(3 marks)

2. (a) ADSU farm is engaged in breeding cows. The cows are fed on various products grown on the farm. Because of the need to ensure certain nutrients, it is necessary to buy additional one or two products, which we shall call A and B. the nutrient constituents (vitamin and minerals) in each unit of the products are given below:

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(3 Marks)

(b) Show the generalized LP model.

(2 Marks)

3. Given the table below:

Base	A	B	C	D	E
1	13	33	31	8	20
2	14	9	17	6	10
3	25	11	12	17	15
4	10	21	13	36	17

Determine the initial basic feasible solution using the least cost method

and North-west corner method.

(3 marks)**(2 marks)**

4. Determine the initial basic feasible solution of the LP model below

$$\text{Max: } Z = 5x_1 + 4x_2$$

$$\text{subject to: } 6x_1 + 4x_2 \leq 10$$

$$x_1 + 2x_2 \leq 6$$

$$-x_1 + x_2 \leq 1$$

$$x_2 \leq 2$$

$$\text{where } x_1, x_2 \geq 0$$

(5 marks)